

indiv_region_increases

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This script test whether non-EUCA per-capita arrest rates rose more than EUCA per-capita rates in all states and AORs and with all denominators

```
aggregate_data <- function(data,
                           cols_to_keep,
                           pop_col,
                           agg_noneuca = TRUE){

  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(aor, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # aggregate regions if requested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA"))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(aor, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # checksum value
  checksum <- data_pop %>%
    group_by(aor, region, pop) %>%
    slice_head(n=1) %>%
    ungroup() %>%
    group_by(region) %>%
    summarize(pop_checksum = sum(pop), .groups = "drop")

  # find geographic keys
  geo_keys <- intersect(cols_to_keep, c("aor", "region"))

  # population map
  pop_map <- data_pop %>%
```

```

    group_by(aor, region, pop) %>%
    slice_head(n = 1) %>%
    ungroup() %>%
    group_by_at(geo_keys) %>%
    summarize(pop = sum(pop), .groups = "drop")

# aggregate arrests, join population back in
data_pop <- data_pop %>%
  group_by_at(cols_to_keep) %>%
  summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
  left_join(pop_map, by = geo_keys) %>%
  mutate(ratio = n_arrests/pop*1000,
         log_pop = log(pop))

# checksum
total_aggregated <- data_pop %>%
  group_by_at(geo_keys) %>%
  summarize(pop = first(pop), .groups = "drop") %>%
  pull(pop) %>%
  sum(na.rm = TRUE)
total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
if (!(dplyr::near(total_checksum, total_aggregated))) {
  warning("Checksum Failed!")
}

return(data_pop)
}

```

MPI, AOR

```

trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv")

## Rows: 113400 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
trajectory_agg <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, aor, and pop.
## i Output is grouped by window, region, and aor.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, aor, pop))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%

```

```

pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
mutate(diff_euca = `0`-`-1`) %>%
mutate(region = "EUCA") %>%
select(diff_euca, aor)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, aor) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)

```

```

##           aor diff_noneuca diff_euca      diff
## 1 San Francisco    4.264320  2.269430  1.994890
## 2  Los Angeles    8.276741  5.120844  3.155898
## 3   Seattle     9.851351  1.485294  8.366057
## 4 New York City  11.419123  2.401267  9.017856
## 5    Boston    12.118928  2.120000  9.998928
## 6  Baltimore   13.466480  3.066667 10.399814
## 7   Newark    16.863535  5.655172 11.208362
## 8   Buffalo   14.383401  2.800355 11.583046
## 9    Dallas   21.485230  9.854304 11.630926
## 10   Chicago  15.004410  3.041237 11.963173
## 11   Atlanta  19.037661  2.491803 16.545858
## 12   San Diego 39.628597 23.037897 16.590700
## 13 Salt Lake City 21.313725  4.000000 17.313725
## 14    Denver   21.175115  3.789474 17.385642
## 15   Houston  25.688052  7.313081 18.374971
## 16    Miami   24.487424  5.859155 18.628269
## 17   Phoenix  26.355172  2.533333 23.821839
## 18  Washington 30.188011  3.791667 26.396344
## 19 Philadelphia 37.107143  5.513514 31.593629
## 20  San Antonio 39.622596  6.199438 33.423158
## 21    St. Paul  42.015075  4.263158 37.751917
## 22  New Orleans 46.310185  4.366667 41.943519
## 23   Detroit   49.284722  2.807692 46.477030
## 24    El Paso  70.740809 10.112767 60.628041
## 25  Harlingen  95.774231  4.821006 90.953224

```

ACS, AOR

```
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv")
```

```

## Rows: 113400 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```
trajectory_agg <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_ACS", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, aor, and pop.
## i Output is grouped by window, region, and aor.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, aor, pop))` for per-operation grouping
## (`?dplyr::dplyr_by`) instead.
```

```
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, aor)
```

```
trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, aor) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

##		aor	diff_noneuca	diff_euca	diff
## 1	San Francisco	1.315308	0.2973953	1.017913	
## 2	Los Angeles	2.191039	0.7735320	1.417507	
## 3	New York City	1.706730	0.2240765	1.482653	
## 4	Seattle	3.356949	0.3009212	3.056027	
## 5	Newark	4.035230	0.5227791	3.512451	
## 6	Boston	4.425621	0.3265075	4.099113	
## 7	Chicago	5.173567	0.5013434	4.672223	
## 8	Baltimore	5.299264	0.4840831	4.815181	
## 9	Miami	6.822478	0.7483105	6.074168	
## 10	Detroit	6.947925	0.2615887	6.686337	
## 11	Dallas	9.519257	1.7399275	7.779330	
## 12	Houston	8.940948	0.9434457	7.997502	
## 13	Salt Lake City	8.712429	0.7096074	8.002822	
## 14	Atlanta	9.164228	0.5172707	8.646957	
## 15	Denver	9.660708	0.7106409	8.950067	
## 16	Phoenix	9.523976	0.2855318	9.238444	
## 17	Philadelphia	10.529848	0.9788445	9.551004	
## 18	Washington	10.350655	0.6527790	9.697876	
## 19	St. Paul	10.619705	0.7666970	9.853008	
## 20	San Diego	13.684053	3.0034199	10.680633	
## 21	Buffalo	12.563828	0.7078944	11.855933	
## 22	San Antonio	21.019638	0.7947887	20.224849	
## 23	New Orleans	21.657867	1.0283866	20.629481	

```
## 24      El Paso      23.714359 1.5316868 22.182672
## 25      Harlingen    30.211050 2.4513559 27.759694
```

```
aggregate_data <- function(data,
                             cols_to_keep,
                             pop_col,
                             agg_noneuca = TRUE){

  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(state, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # aggregate regions if requested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA"))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(state, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # checksum value
  checksum <- data_pop %>%
    group_by(state, region, pop) %>%
    slice_head(n=1) %>%
    ungroup() %>%
    group_by(region) %>%
    summarize(pop_checksum = sum(pop), .groups = "drop")

  # find geographic keys
  geo_keys <- intersect(cols_to_keep, c("state", "region"))

  # population map
  pop_map <- data_pop %>%
    group_by(state, region, pop) %>%
    slice_head(n = 1) %>%
    ungroup() %>%
    group_by_at(geo_keys) %>%
    summarize(pop = sum(pop), .groups = "drop")

  # aggregate arrests, join population back in
  data_pop <- data_pop %>%
    group_by_at(cols_to_keep) %>%
    summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
    left_join(pop_map, by = geo_keys) %>%
```

```

    mutate(ratio = n_arrests/pop*1000,
           log_pop = log(pop))

# checksum
total_aggregated <- data_pop %>%
  group_by_at(geo_keys) %>%
  summarize(pop = first(pop), .groups = "drop") %>%
  pull(pop) %>%
  sum(na.rm = TRUE)
total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
if (!(dplyr::near(total_checksum, total_aggregated))) {
  warning("Checksum Failed!")
}

return(data_pop)
}

```

MPI, state

```
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI.csv")
```

```

## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
trajectory_agg <- aggregate_data(trajectory, c("region", "state", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, state, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, state, and pop.
## i Output is grouped by window, region, and state.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, state, pop))` for per-operation
## grouping (`?dplyr::dplyr_by`) instead.

trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, state)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, state) %>%

```

```
merge(trajjectory_agg_euca) %>%
mutate(diff = diff_noneuca - diff_euca) %>%
arrange(diff)
```

##	state	diff_noneuca	diff_euca	diff
## 1	OKLAHOMA	0.3168317	0.000000	0.3168317
## 2	CONNECTICUT	3.7443182	1.363636	2.3806818
## 3	HAWAII	6.6428571	3.928571	2.7142857
## 4	CALIFORNIA	9.4052990	5.367521	4.0377776
## 5	ILLINOIS	8.6833013	2.909091	5.7742104
## 6	RHODE ISLAND	7.8518519	2.000000	5.8518519
## 7	ALASKA	8.4444444	1.500000	6.9444444
## 8	WASHINGTON	8.6195286	1.276596	7.3429329
## 9	NORTH CAROLINA	10.9014085	2.541667	8.3597418
## 10	NEVADA	13.5645161	4.500000	9.0645161
## 11	NEW YORK	12.8112450	2.545455	10.2657904
## 12	MARYLAND	14.3631285	3.266667	11.0964618
## 13	OREGON	12.9926471	1.882353	11.1102941
## 14	WISCONSIN	14.2000000	3.000000	11.2000000
## 15	NEW JERSEY	16.8590604	5.620690	11.2383707
## 16	MASSACHUSETTS	14.6676301	2.428571	12.2390586
## 17	COLORADO	17.5188679	2.611111	14.9077568
## 18	FLORIDA	24.5585429	5.859155	18.6993880
## 19	KANSAS	21.7088608	2.750000	18.9588608
## 20	INDIANA	25.1428571	5.500000	19.6428571
## 21	DELAWARE	19.6785714	-1.000000	20.6785714
## 22	GEORGIA	23.9232456	2.750000	21.1732456
## 23	KENTUCKY	25.6964286	3.600000	22.0964286
## 24	VIRGINIA	26.5176471	3.950000	22.5676471
## 25	ARIZONA	26.4068966	2.533333	23.8735632
## 26	TEXAS	34.6857440	9.177778	25.5079662
## 27	UTAH	28.8740157	3.300000	25.5740157
## 28	VERMONT	27.0000000	1.000000	26.0000000
## 29	PENNSYLVANIA	31.6018519	5.235294	26.3665577
## 30	SOUTH CAROLINA	28.7322835	1.846154	26.8861296
## 31	NEBRASKA	31.6274510	4.000000	27.6274510
## 32	IOWA	34.5454545	4.400000	30.1454545
## 33	IDAHO	35.7272727	5.500000	30.2272727
## 34	LOUISIANA	37.3440860	1.000000	36.3440860
## 35	NEW HAMPSHIRE	40.3750000	3.000000	37.3750000
## 36	MISSOURI	40.7894737	1.875000	38.9144737
## 37	ARKANSAS	43.5967742	3.750000	39.8467742
## 38	TENNESSEE	45.3431953	3.545455	41.7977407
## 39	OHIO	47.6172840	5.400000	42.2172840
## 40	MINNESOTA	52.7142857	5.500000	47.2142857
## 41	MICHIGAN	51.1587302	1.062500	50.0962302
## 42	MISSISSIPPI	61.7826087	8.000000	53.7826087
## 43	DISTRICT OF COLUMBIA	63.3461538	2.500000	60.8461538
## 44	ALABAMA	71.2121212	8.600000	62.6121212
## 45	NEW MEXICO	70.9333333	2.000000	68.9333333
## 46	MONTANA	131.0000000	2.000000	129.0000000
## 47	NORTH DAKOTA	208.0000000	3.000000	205.0000000
## 48	MAINE	364.0000000	4.000000	360.0000000
## 49	SOUTH DAKOTA	545.0000000	10.000000	535.0000000

```
## 50          WYOMING  840.0000000 22.000000  818.0000000
## 51      WEST VIRGINIA 1973.0000000 25.000000 1948.0000000
```

ACS, state

```
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI.csv")
```

```
## Rows: 231336 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
trajectory_agg <- aggregate_data(trajectory, c("region", "state", "window"), "pop_ACS", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, state, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, state, and pop.
## i Output is grouped by window, region, and state.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, state, pop))` for per-operation
## grouping (`?dplyr::dplyr_by`) instead.
```

```
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, state)
```

```
trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, state) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

##	state	diff_noneuca	diff_euca	diff
## 1	HAWAII	0.8680637	1.17370892	-0.3056453
## 2	OKLAHOMA	0.1429044	0.00000000	0.1429044
## 3	CONNECTICUT	1.5634267	0.22448369	1.3389430
## 4	ALASKA	1.7696642	0.42298202	1.3466822
## 5	VERMONT	1.8634826	0.07418948	1.7892932
## 6	CALIFORNIA	2.7408858	0.70138680	2.0394990
## 7	NEW YORK	2.5608119	0.29397329	2.2668386
## 8	WASHINGTON	2.7400603	0.25381033	2.4862500
## 9	ILLINOIS	3.1196902	0.50043527	2.6192549
## 10	RHODE ISLAND	3.2869746	0.23938478	3.0475898

## 11	WISCONSIN	3.4641046	0.33424322	3.1298614
## 12	NEW JERSEY	4.0341597	0.51959147	3.5145683
## 13	NEVADA	4.7216774	0.73766864	3.9840088
## 14	MASSACHUSETTS	5.1681771	0.40071185	4.7674652
## 15	OREGON	5.4246217	0.37623157	5.0483901
## 16	NEW HAMPSHIRE	5.6295315	0.52230231	5.1072292
## 17	MARYLAND	5.6521084	0.51565378	5.1364547
## 18	NORTH CAROLINA	5.7765505	0.50840536	5.2681451
## 19	MICHIGAN	5.9066735	0.10509397	5.8015795
## 20	FLORIDA	6.8422926	0.74831046	6.0939821
## 21	DELAWARE	5.9228206	-0.17146776	6.0942884
## 22	NORTH DAKOTA	7.4309600	0.43975374	6.9912062
## 23	INDIANA	8.3868490	0.91975167	7.4670974
## 24	COLORADO	8.0669509	0.48859596	7.5783550
## 25	OHIO	8.1063131	0.46034236	7.6459708
## 26	KENTUCKY	8.4504892	0.61393636	7.8365528
## 27	PENNSYLVANIA	8.8398734	0.94148013	7.8983933
## 28	KANSAS	8.9241112	0.57830819	8.3458030
## 29	VIRGINIA	9.0278174	0.65019506	8.3776224
## 30	IOWA	9.5709446	0.92514718	8.6457974
## 31	ARIZONA	9.5426672	0.28553180	9.2571354
## 32	GEORGIA	10.3189227	0.57255882	9.7463639
## 33	MONTANA	10.0421617	0.16283993	9.8793218
## 34	MINNESOTA	11.1202808	0.95613928	10.1641415
## 35	MISSOURI	10.8048573	0.27411780	10.5307395
## 36	NEBRASKA	11.5422871	0.68440414	10.8578830
## 37	MAINE	11.9504908	0.17265193	11.7778389
## 38	TEXAS	13.8881945	1.39670472	12.4914897
## 39	IDAHO	13.5812282	1.01033295	12.5708952
## 40	UTAH	15.3334727	0.71953427	14.6139385
## 41	SOUTH CAROLINA	15.5302369	0.40959126	15.1206457
## 42	SOUTH DAKOTA	18.5475088	1.71408982	16.8334190
## 43	LOUISIANA	17.4556823	0.20524398	17.2504383
## 44	ARKANSAS	19.5532343	1.50142636	18.0518079
## 45	TENNESSEE	21.2641972	0.75773767	20.5064595
## 46	MISSISSIPPI	23.1339031	1.75054705	21.3833561
## 47	DISTRICT OF COLUMBIA	22.9784028	0.55859680	22.4198060
## 48	NEW MEXICO	24.0451977	0.21640338	23.8287944
## 49	ALABAMA	28.5308437	1.57480315	26.9560405
## 50	WYOMING	55.1144938	4.29435877	50.8201350
## 51	WEST VIRGINIA	86.4062363	3.25478453	83.1514518